



BIG DATA ANALYTICS IN FINANCE

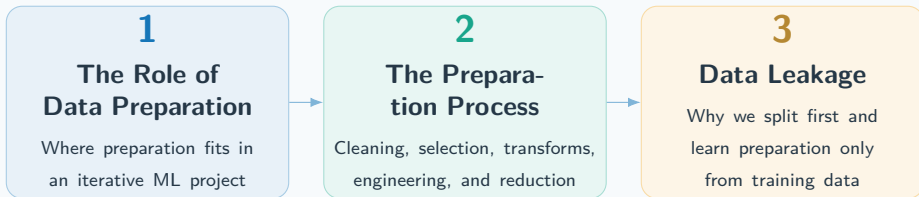
Foundations of Data Preparation for Machine Learning

Project lifecycle, preparation tasks, and data leakage

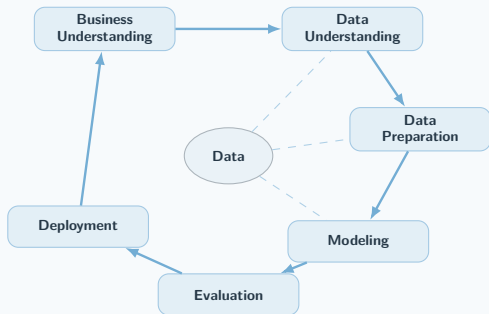
Dr. Nguyen Tuan Long National Economics University | 2026



Agenda



1. The Role of Data Preparation



An iterative project cycle

A machine-learning project is not a straight line. Business understanding, data understanding, preparation, modeling, evaluation, and deployment continually inform one another.

Practical implication

Model errors often send us back to preparation to revise cleaning, transformations, or features.

1. The Role of Data Preparation in the Pipeline



Where preparation operates

Multiple sources are selected and merged into raw data. Cleaning and transformation convert those records into model-ready features.

Core message

Data preparation is the bridge between collected data and credible insight.

1. Four Steps in a Machine-Learning Project

Step 1: Define the Problem

Understand the problem, collect relevant data, and choose an evaluation metric.

Step 2: Prepare the Data

Clean, transform, and select features. This is the foundation for later steps.

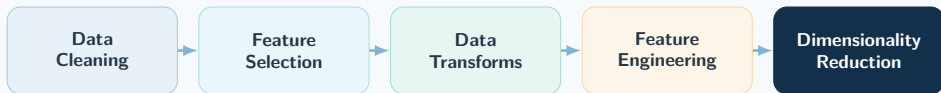
Step 3: Evaluate Algorithms

Test multiple models and estimate generalization with cross-validation.

Step 4: Finalize the Model

Train the selected workflow, validate it, and deploy it for operational use.

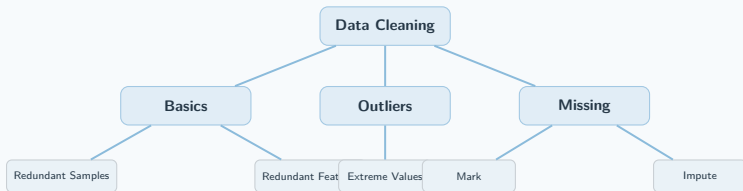
2. The Data Preparation Process



Five main task groups

Cleaning improves validity; selection removes unnecessary inputs; transforms change representation; engineering creates new information; dimensionality reduction creates a compact representation.

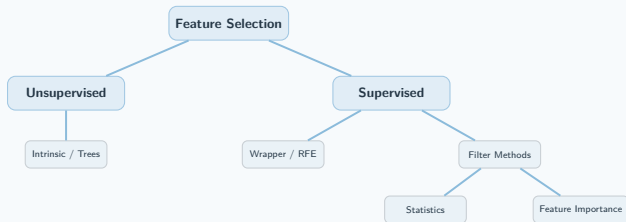
2.1. Data Cleaning



Purpose

Correct invalid, redundant, extreme, or missing observations before modeling.

2.2. Feature Selection



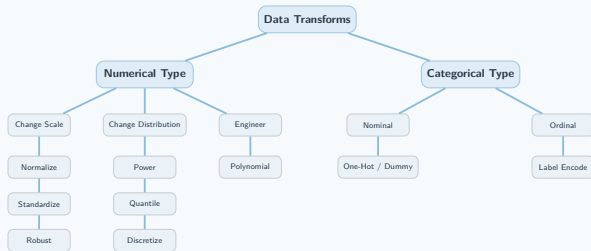
Purpose

Retain informative predictors while removing irrelevant or redundant variables.

Method families

Intrinsic, wrapper, and filter methods.

2.3. Data Transforms



Numerical data

Change scale, distribution, or representation.

Categorical data

Encode nominal and ordinal variables appropriately.

2.4. Feature Engineering

Purpose

Create new features from existing data so a model can represent useful relationships more directly.

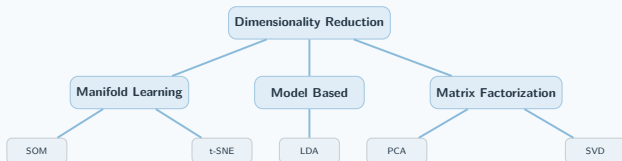
Original examples

- ▶ Split a date into day, month, and year.
- ▶ Combine population and area into population density.

Finance and business examples

- ▶ Debt-to-income ratio.
- ▶ Transaction velocity within rolling windows.
- ▶ Returns and volatility from market prices.
- ▶ Customer tenure from account-opening dates.

2.5. Dimensionality Reduction



Technique families

- ▶ Manifold learning
- ▶ Model-based reduction
- ▶ Matrix factorization

Goal

Compress many variables into fewer informative dimensions.

3. Data Leakage

Data leakage occurs when information from the test set enters the model-training process.

- ▶ Test statistics influence scaling or imputation.
- ▶ Future information appears in historical predictors.
- ▶ Duplicate entities occur in both training and testing data.

The most important concept

Leakage can make validation performance look excellent while producing a model that is completely useless in practice.

3. Correct vs. Incorrect Process

Incorrect process

1. Take the entire dataset.
2. Prepare it, for example by scaling.
3. Split into training and test sets.
4. Train the model.

Problem

Minima, maxima, means, or other test-set information have already leaked into the preparation step.

Correct process

1. **Split** raw data first.
2. **Fit** preparation only on training data.
3. **Transform** training and test data with the fitted object.
4. Train and evaluate on untouched test observations.

Golden rule

SPLIT FIRST, PREPARE LATER.

Summary: Key Takeaways

Iterative foundation

Data preparation is an iterative process and the foundation of a machine-learning project.

Five task groups

Cleaning, feature selection, transforms, engineering, and dimensionality reduction.

Leakage prevention

Always split data before learning preparation parameters.

Best practice

Use a Scikit-learn Pipeline to connect preparation and modeling in one reproducible workflow.

DATA PREPARATION FOUNDATIONS

Q&A

Questions and discussion